

Skills Summary

Zeros, Multiplicity, and Graph Behavior

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading zeros from factored form

Rule:

If $P(x) = a(x - r_1)^{m_1}(x - r_2)^{m_2} \cdots$, then the zeros are $x = r_1, r_2, \dots$, and the exponent on a factor is that zero's **multiplicity**. Set each factor equal to zero and solve; a factor $(x + k)$ gives the zero $x = -k$.

Example: Find the zeros of $f(x) = (x - 5)(x + 2)^2$.

Step 1. The polynomial is already a product, and a product is zero exactly when one of its factors is zero, so no multiplying out is needed.

Step 2. $x - 5 = 0$ gives $x = 5$; $x + 2 = 0$ gives $x = -2$, and the exponent 2 says that zero has multiplicity two.

Zeros: **5** and **-2**

Try it: Find the zeros of $g(x) = (x - 1)(x + 6)^2$.

check: 1 and 6

Skill 2 — Factor first, then read the zeros

Method: pull out the greatest common factor, factor what is left, then set each factor to zero. A common factor of x always produces the zero $x = 0$, which is also the x -intercept at the origin.

Example: Factor $h(x) = x^3 - 6x^2 + 8x$ and list its zeros.

Step 1. Every term contains an x , so take out the greatest common factor before trying anything else — that turns a cubic into a quadratic you already know.

Step 2. $x^3 - 6x^2 + 8x = x(x^2 - 6x + 8)$, and two numbers with product 8 and sum -6 are -2 and -4 .

$h(x) = x(x - 2)(x - 4)$, so the zeros are **0, 2, 4**

Try it: Factor $p(x) = x^3 - x^2 - 12x$ and list its zeros.

check: 0, 4, -3

Skill 3 — Multiplicity and what the graph does

At a zero of multiplicity m :

- m odd — the curve **crosses** the x -axis there;
- m even — the curve **touches** the axis and turns back;
- $m \geq 3$ — it also flattens out as it goes through or turns.

Far left and far right are decided by the leading term alone: odd degree sends the two ends opposite ways, even degree sends them the same way, and a negative leading coefficient flips both.

Example: For $f(x) = (x + 1)^2(x - 3)$, say what happens at each zero, and find $f(0)$.

Step 1. The exponent on each factor is its multiplicity, and multiplicity is the only thing that decides touching versus crossing.

Step 2. $x = -1$ comes from a squared factor, so the curve touches and turns there; $x = 3$ comes from a single factor, so the curve crosses. The y -intercept is $f(0) = (0 + 1)^2(0 - 3)$.

$$f(0) = -3$$

Try it: For $g(x) = (x - 2)^2(x + 4)$, which zero is only touched, and what is $g(0)$?

check: $g(0) = 8$; $x = 2$ is touched

Skill 4 — Building a polynomial from its zeros

Rule: a zero $x = r$ of multiplicity m contributes the factor $(x - r)^m$. Multiply the factors together, then multiply by a constant a if the graph must also pass through a stated point. The degree is the sum of the multiplicities.

Example: Write a polynomial with zeros $x = 2$ (multiplicity one) and $x = -3$ (multiplicity two), leading coefficient 1, then expand it.

Step 1. Each zero becomes a factor and its multiplicity becomes that factor's exponent, so the factored form can be written straight down.

Step 2. $(x - 2)(x + 3)^2 = (x - 2)(x^2 + 6x + 9)$, and multiplying out term by term gives $x^3 + 6x^2 + 9x - 2x^2 - 12x - 18$.

$$x^3 + 4x^2 - 3x - 18$$

Try it: Write and expand a polynomial with zeros $x = -1$ (multiplicity one) and $x = 5$ (multiplicity two), leading coefficient 1.

check: $x^3 + 9x^2 + 14x - 50$